



MENAI HIGH SCHOOL

ASSESSMENT TASK NOTIFICATION



Computing Technology - Assessment Task 3

Teacher: A.Kasper

Due Date: Term 3, Week 5

Task Description

Your challenge is to design and build a real database app using Google Sheets and Google AppSheet. Your app should solve a real problem, organise useful information, and give users a simple way to add, search, edit and view data. Instead of writing code from scratch, you will use AppSheet to create a professional no-code application connected to a structured Google Sheets database. You will define the problem, research similar systems, and plan your solution using wireframes and flowcharts. After developing the system, you will test and evaluate its functionality, ensuring data is handled securely and ethically. Your final submission will include your working program, documentation of your process, and a presentation demonstrating your solution.

Scenario	What the app could do
1. Collector Hub	Track a collection such as Pokemon cards, sneakers, books, vinyl, games or sports cards. Include categories, rarity/value, images and wish-list items.
2. Canteen Click & Collect	Create an ordering app for a school canteen with menu items, customer orders, order status and daily summaries.
3. Sport Team Tracker	Track players, training attendance, positions, game statistics and awards.
4. Study Quest	Create a study planner with subjects, tasks, due dates, priority levels and progress tracking.
5. Entertainment Booking App	Manage bookings for music, sport, e-sports, theatre, movie nights or lunchtime activities. Organise a school event with jobs, student volunteers, equipment, timeslots and check-in status

The following points must be completed within your submission:

Identifying and Defining

Each group will select one of the four project scenarios and define the problem the chosen system will solve. The following points must be addressed within this section:

- **Define** your project scenario, ensuring you clearly identify the specific problem the system will solve. *(Outcome CT5-THI-01)*
- **Identify** the key user roles (e.g. teacher, student, admin staff) and their specific responsibilities. *(Outcome CT5-THI-01)*
- **Describe** the types of data the system will need to process, such as awards, orders, event details, or PD forms, and the security measures needed. **Explain** how the system will securely store, transmit, and manage this data. *(Outcome CT5-DAT-01)*
- **Outline** the ethical and security implications of your system, particularly how data privacy will be protected and responsible technology use ensured. *(Outcome CT5-SAF-01)*

Researching and Planning

Each group will need to research existing systems similar to their chosen system and create a detailed plan for the development of your system. This includes developing diagrams, wireframes, and flowcharts to illustrate your design.

- Research and investigate existing systems similar to your project (e.g. online order systems, awards management) and **analyse** their features to inform your design. *(Outcome CT5-DPM-01)*
- **Create** wireframes and flowcharts for your system's design, ensuring the user interface (UI) and user experience (UX) meet user needs. *(Outcome CT5-DES-01)*
- **Plan** how your system will manage and store data, identifying whether it will use databases, cloud storage, or other methods. *(Outcome CT5-DAT-02)*
- **Assign** group roles (e.g., Lead Developer, UI/UX Designer, Project Manager) and set a timeline for project development to ensure each member contributes effectively. *(Outcome CT5-COL-01)*

Producing and Implementing

Each group will need to start coding and implementing their system. Each group will need to

apply their designs and ensure that the system operates securely, efficiently, and meets user needs.

- **Develop** the code for your application, ensuring that user authentication, secure data handling, and user interface features are implemented effectively. (*Outcome CT5-DPM-01*)
- **Ensure** your system correctly handles data such as awards, canteen orders, PD submissions, or entertainment details. (*Outcome CT5-DAT-02*)
- Implement necessary accessibility and security features (e.g. password protection, encryption) to safeguard user data and meet ethical standards. (*Outcome CT5-SAF-01*)

Minimum App requirements

Your finished AppSheet app must include:

- A Google Sheets database with at least three connected tables, for example: Users, Items, Orders/Bookings/Tasks.
- A unique key field for each table, such as ItemID, OrderID or UserID.
- Appropriate data types: text, number, date, time, image, email, yes/no, enum/dropdown and reference fields where suitable.
- At least four AppSheet views, such as deck, table, detail, form, calendar, gallery, chart or dashboard.
- At least one form that lets a user add new data.
- At least one search, filter or slice so users can find important records quickly.
- At least one validation rule or dropdown list to reduce data entry mistakes.
- Evidence of security and responsible data choices, such as limiting personal information, using sharing settings carefully and explaining who should have access.
- Accessibility features, such as clear labels, readable layout, logical navigation and colour choices that are not the only way information is communicated.
- Testing evidence showing that the app was improved after feedback

Testing and Evaluating

Each group will be required to test the system with real-world data and evaluate how well it meets the original design goals. The following points must be addressed within this section:

- **Test** the functionality of your system using real or mock data, verifying that it handles user roles correctly and secures data properly. **Explain** how data is stored, transmitted, and secured during the testing phase. (*Outcome CT5-DAT-01*)
- **Document** any issues or bugs encountered during testing and iterate on your solution to

make necessary fixes. (*Outcome CT5-DPM-01*)

- **Evaluate** whether your system meets the original design goals, including usability, performance, and security. (*Outcome CT5-DES-01*)
- **Reflect** on the entire process, including your collaborative efforts, and document how each team member contributed to the final outcome. (*Outcome CT5-COL-01*)

Submission Requirements:

Submit documentation that includes your group's collaborative work, problem definition, design, and system testing.

- **Document** and **explain** individual and group contributions throughout the project. (*Outcome CT5-COL-01*)
- Provide a working version of your coded program that addresses the project scenario.
- **Demonstrate** an iterative approach in the development of the system. (*Outcome CT5-DPM-01*)
- Ensure the system's design is user-friendly and meets the desired functionality. (*Outcome CT5-DES-01*)
- Effectively communicate ideas, processes, and solutions using appropriate media such as flowcharts or presentations. (*Outcome CT5-COM-01*)

**** To ensure that you get the best results possible, make sure that you are completing all areas of the assessment matrix ****

Submission Details:

- The task will be completed both in class and at home.
- All word-processed files must be saved as a pdf and all computer files must be compressed before uploading it to Google Classroom by the submission time on the due date. An E-Grade will be awarded for late submission.

**** Please note that where plagiarism has occurred, students will receive an E-Grade automatically and be asked to complete the task again. ****

Outcomes Assessed

CT5-SAF-01	Selects and applies safe, secure, and responsible practices in the ethical use of data and computing technology
CT5-DPM-01	Applies iterative processes to define problems and plan, design, develop and evaluate computing solutions
CT5-COL-01	Manages, documents, and explains individual and collaborative work practices
CT5-DAT-01	Explains how data is stored, transmitted, and secured in digital systems and how information is communicated in a range of contexts
CT5-COM-01	Communicates ideas, processes and solutions using appropriate media
CT5-THI-01	Applies computational, design and systems thinking to the development of computing solutions
CT5-DAT-02	Acquires, represents, analyses, and visualises simple and structured data
CT5-DES-01	Designs and creates user interfaces and the user experience

Done	Item
☐	Google Sheets database with named tables and sensible fields
☐	Working Google AppSheet app link
☐	Project documentation saved as a PDF
☐	Research comparing at least two existing apps or systems
☐	Wireframes for at least three screens
☐	Flowchart showing how a user completes one important task
☐	Explanation of group roles
☐	Testing table with bugs, fixes and retesting evidence
☐	Group contribution log and individual reflection

	A	B	C	D	E
Define your project scenario, ensuring you clearly identify the specific problem the system will solve. (Outcome CT5-THI-01)	Clearly defines the problem and explains in depth how the system solves it. The scenario is specific and well-aligned with the needs of users.	Defines the problem with adequate detail but lacks some depth or clarity in the explanation of how the system solves it.	Defines the problem but with insufficient detail or clarity on how the system will solve the issue. The solution may not be fully clear.	Defines the problem vaguely or with minimal relevance to the chosen system scenario. It may lack alignment with the user needs.	Fails to define the problem, leaving unclear what the system is meant to address. No clear connection to the user or their needs.
Identify the key user roles (e.g. teacher, student, admin staff) and their specific responsibilities. (Outcome CT5-THI-01)	Identifies all relevant roles (e.g., teacher, student, admin) and clearly defines the responsibilities and interactions of each role in the system. Roles are well articulated with clear, purposeful relationships.	Identifies the key user roles and outlines responsibilities. Some roles and interactions may lack clarity or are not fully defined.	Identifies some roles but does not fully define their responsibilities or interactions. Some roles may be missing.	Mentions user roles but does not clearly define responsibilities or their interactions with the system.	No clear identification or definition of key roles. Responsibilities are not defined or are completely missing.
Describe the types of data the system will need to process, such as awards, orders, event details, or PD forms, and the security measures needed. Explain how the system will securely store, transmit, and manage this data. (Outcome CT5-DAT-01)	Provides a comprehensive explanation of the data types the system will process (e.g., awards, orders, forms). The explanation covers secure data handling, storage, encryption, and transmission methods. Security measures are thorough and detailed.	Provides an adequate explanation of the data types and basic security measures (e.g., encryption, secure storage), but lacks some depth in the explanation of security features or their application.	Mentions data processing and security but provides only basic details. Lacks clear explanation of how data will be securely handled, stored, and transmitted.	Provides minimal information about data processing or security. Key concepts are missing, leaving gaps in the understanding of how data will be handled.	Does not explain how data will be handled or stored securely. No security measures are mentioned.
Outline the ethical and security implications of your system, particularly how data privacy will be protected and responsible technology	Thorough discussion of data privacy, responsible technology use, and any ethical concerns. A deep reflection on how the system ensures responsible use of data and its impact on users.	Provides a reasonable overview of ethical concerns, including data privacy and responsible technology use, but lacks full depth or detail on	Mentions ethical or security issues, but the explanation is vague or lacks clear focus on data privacy or responsible use of technology.	Minimal or no discussion on ethical or security concerns. There is little attention given to how data privacy or responsible use will be ensured.	No mention of ethical concerns or data privacy. No consideration of responsible technology use.

use ensured. (Outcome CT5-SAF-01)		how these issues are addressed in the system.			
Research and investigate existing systems similar to your project (e.g. online order systems, awards management) and analyse their features to inform your design. (Outcome CT5-DPM-01)	Conducts comprehensive research on similar systems, analyzing key features, strengths, and weaknesses. Demonstrates how this research informs and influences design decisions.	Conducts adequate research, analyzing key features and drawing conclusions about design, but may lack depth in analysis or critical evaluation.	Conducts minimal or basic research, which lacks critical analysis. May not fully consider how it informs the design decisions.	Minimal or no research conducted. Research may not be relevant, lacking any direct connection to the project design.	No research on existing systems conducted.
Create wireframes and flowcharts for your system's design, ensuring the user interface (UI) and user experience (UX) meet user needs (Outcomes CT5-DES-01)	Well-organized, clear wireframes and flowcharts that illustrate the system's UI/UX effectively. Includes details like layout, user interactions, and navigational flow.	Wireframes and flowcharts are presented with adequate detail but may lack clarity in some areas. UI/UX is represented but could be more refined or detailed.	Basic wireframes and flowcharts are included but may lack essential details or clarity. UI/UX is underdeveloped.	Wireframes or flowcharts are poorly developed or missing significant aspects, such as user interaction details or UI structure.	No wireframes or flowcharts are submitted.
Plan how your system will manage and store data, identifying whether it will use databases, cloud storage, or other methods. (Outcome CT5-DAT-02)	Clear and thorough explanation of how data will be managed, stored (e.g., database, cloud storage), and processed. Provides detailed plans for data backups, security, and scalability.	Provides a solid plan for data management but may lack specific details or considerations for data backups, security, or scalability.	Provides an unclear or incomplete plan for data management, lacking details about storage or processing methods.	No clear data management plan provided. There are significant gaps in how data will be stored or processed.	No data management plan is provided.
Assign group roles (e.g., Lead Developer, UI/UX Designer, Project Manager) and set a timeline for project development to ensure each member contributes	Roles and responsibilities are clearly defined and align with the skills of each group member. A detailed, realistic timeline is set for each phase of the project, with clear deadlines.	Roles are defined, but there may be minor gaps in matching responsibilities to skills. The timeline is set but may lack some specificity or clarity in how each phase will be completed.	Roles are loosely defined or unclear, and the timeline is underdeveloped. It may not be realistic or lack sufficient detail for managing the project effectively.	Roles and timeline are vaguely defined, and there is minimal alignment with the needs of the project. The timeline is unrealistic or not practical.	No clear roles or timeline outlined. Project management is unclear or non-existent.

effectively. (Outcome CT5-COL-01)					
Develop the code for your application, ensuring that user authentication, secure data handling, and user interface features are implemented effectively. (Outcome CT5-DPM-01)	Code is well-organized, clean, and efficient. Implements user authentication, secure data handling, and robust UI features. Code is fully documented and easy to follow.	Code is well-structured but may have minor inefficiencies or missing features. Some aspects of user authentication or data handling may need refinement.	Code is functional but may be disorganized or inefficient. User authentication or data handling may be incomplete or improperly implemented.	Code is incomplete or poorly structured, with significant issues in user authentication, data handling, or core functionality.	Code is fundamentally flawed, with major issues preventing the system from functioning as intended.
Ensure your system correctly handles data such as awards, canteen orders, PD submissions, or entertainment details. (Outcome CT5-DAT-02)	Data is handled correctly according to design, with secure storage, encryption, and transmission methods fully implemented.	Data handling is mostly correct but may have minor issues with data storage or transmission. Security features may be less robust.	Data handling is incomplete or insecure. Security measures may be lacking or improperly implemented.	Data handling is ineffective or insecure. Major flaws exist in how data is stored or transmitted.	No data handling implemented.
Implement necessary accessibility and security features (e.g. password protection, encryption) to safeguard user data and meet ethical standards (Outcome CT5-SAF-01)	Fully implements security (e.g., password protection, encryption) and accessibility features (e.g., screen reader support, keyboard navigation) that safeguard user data.	Implements necessary security and accessibility features but may lack some minor improvements or refinements.	Implements basic security and accessibility features, but there are significant gaps that could hinder accessibility or data security.	Security and accessibility features are missing or incorrectly implemented. These flaws can expose user data or make the system hard to use for some users.	No security or accessibility features implemented.
Test the functionality of your system using real or mock data, verifying that it handles user roles correctly and secures data properly. Explain how data is stored, transmitted, and secured	Comprehensive testing with real or mock data. All major bugs identified and fixed. Proper data handling, security, and functionality are verified during testing. Explanation of data storage, transmission,	Adequate testing with mock data. Most bugs identified and fixed, but some minor issues remain. Security and data handling mostly correct, with slight gaps. The explanation of how data	Basic testing with mock data. Some bugs remain unresolved, but the system functions as intended overall. Data storage and security are partially explained, with some gaps.	Inadequate testing. Major bugs or issues are not identified or resolved. Testing is insufficient to verify key system functions. Data handling and security	No testing conducted. No explanation provided on how data is stored, transmitted, or secured.

during the testing phase. (Outcome CT5-DAT-01)	and security is clear and thorough.	is handled is provided but lacks depth.		are barely explained or unclear.	
Document any issues or bugs encountered during testing and iterate on your solution to make necessary fixes. (Outcome CT5-DPM-01)	Thoroughly documents all issues or bugs encountered during testing. Iterates on the solution effectively and makes necessary fixes. Demonstrates a clear process of bug resolution and improvement.	Documents most issues and bugs encountered, but some minor issues may be missed. Iterates on solutions and makes fixes, but some fixes could be more thorough.	Documents some issues but may miss a few or fail to address all the bugs. Iterates on some solutions, but the fixes may not be fully effective.	Limited documentation of issues and fixes. Many bugs remain unaddressed. The iterative approach to problem-solving is either missing or poorly executed.	No documentation of issues or bugs. No iteration on solutions or fixes provided.
Evaluate whether your system meets the original design goals, including usability, performance, and security. (Outcome CT5-DES-01)	Comprehensive evaluation of system functionality, usability, performance, and security. The system fully meets original design goals and user needs. Detailed analysis of strengths and weaknesses with recommendations for improvement.	Evaluates the system's functionality, usability, and security, but may miss minor aspects of performance or security. The system mostly meets design goals with minor weaknesses.	Evaluation is basic or incomplete, focusing on only some aspects of the system. The system meets some design goals but has clear performance, usability, or security gaps.	Evaluation is superficial or incomplete, with little analysis of how the system meets design goals. The system fails to meet key design goals or user needs.	No meaningful evaluation. The system does not meet the original design goals or user needs.
Reflect on the entire process, including your collaborative efforts, and document how each team member contributed to the final outcome. (Outcome CT5-COL-01)	Clear and thoughtful reflection on individual and team contributions. Each member's role is well-documented, and the collaborative process is fully evaluated. Insightful analysis of how teamwork influenced the final outcome.	Reflects on individual and team contributions, but with minor gaps in evaluating the collaboration process. Some members' roles are not clearly defined.	Minimal reflection on team contributions. Focuses mainly on individual effort, with little insight into how the team worked together.	Little to no reflection on team contributions or collaboration. Minimal or no insight into how teamwork influenced the outcome.	No reflection on the process or contributions. No documentation of how the team worked together.

